💻 □ Metropolitan University, Sylhet

Department of Computer Science & Engineering

Lab Report – 02:

Greedy Algorithm(Activity Selection)

Course Title: Algorithm Design and Analysis Lab

Course Code: CSE 232

Submitted To:

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Batch: CSE 62nd (A)

Date of Submission:

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problem:

Greedy Algorithm(Activity Selection)

CODE :

```
// Activity Selection Problem using Greedy Method
// Name: Mahdi Hasan Shuvo | ID: 251-115-030 | Batch: CSE 62nd (A)

#include<iostream>
#include<algorithm>
#include<vector>
using namespace std;

struct Activity
{
    int starting;
    int ending;
};


bool compare(Activity a, Activity b)
{
    return a.ending < b.ending;
}

int main()
{
    int n;
    cout << "Enter activity number: ";
    cin >> n;

    vector <Activity> acts(n);

    for (int i = 0; i < n; i++)
    {
        cout << "Start time of activity-" << i+1 << ": ";
        cin >> acts[i].starting;
        cout << "End time of activity-" << i+1 << ": ";
        cin >> acts[i].ending;
    }

    sort(acts.begin(), acts.end(), compare);

    int selected = 0;
```

```cpp
    int last_end = 0;
    last_end = acts[0].ending;
    selected = 1;
    for (int i = 1; i < n; i++)
    {
        if (acts[i].starting >= last_end)
        {
            selected++;
            last_end = acts[i].ending;
        }
    }

    cout << "Maximum possible activity is " << selected << endl;

    return 0;
}
```

Output Output Output

OUTPUT: